

Bound sets for biobjective combinatorial optimization problems

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Abstract

In this paper we introduce the concept of bound sets for multiobjective discrete optimization. We prove general results on lower and upper bound sets for combinatorial optimization problems with multiple objectives. We present general algorithms for constructing lower and upper bound sets for biobjective problems and provide numerical results on five different problem types.

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1. Introduction

1.1. Multiobjective combinatorial optimization

Multiobjective combinatorial optimization problems can be formulated as follows:

$$\min\{Cx : Ax \geq b, x \in \mathbb{Z}^n\}. \quad (\text{MOCO})$$

Here, C is a $p \times n$ objective function matrix, where c^k denotes the k th row of C . A is a $m \times n$ matrix of constraint coefficients and $b \in \mathbb{R}^m$. Usually the entries of C , A , and b are integers. The feasible set $X := \{Ax \geq b, x \in \mathbb{Z}^n\}$ may describe a combinatorial structure such as e.g. spanning trees of a graph, paths, matchings etc. We shall assume that X is a finite set. By $Y = CX$ we denote the image of X under C in $\mathbb{R}^p$.

A feasible solution $\hat{x} \in X$ is called efficient (or Pareto optimal) if there does not exist $x \in X$ such that $c^k x \leq c^k \hat{x}$ for all $k = 1, \dots, p$ and $c^k x < c^k \hat{x}$ for some k , $1 \leq k \leq p$. In other words, no solution is at least as good as $\hat{x}$ for all objectives and strictly better for at least one.

Efficiency refers to solutions x in decision space. In terms of the criterion space, with objective vectors $Cx \in \mathbb{R}^p$, we use the notion of nondominance: If x is efficient then $Cx = (c^1 x, \dots, c^p x)$ is called nondominated (or also efficient). The set of efficient solutions is $X_E \subset X$, the set of nondominated objective vectors (or points) is $Y_N \subset Y$. Surveys of the state of the art in multiobjective combinatorial optimization are available in [1,2].

For $y^1, y^2 \in \mathbb{R}^p$ we shall write $y^1 \leq y^2$ if $y_k^1 \leq y_k^2$, $k = 1, \dots, p$, $y^1 \leq y^2$ if $y^1 \leq y^2$ but $y^1 \neq y^2$ and $y^1 < y^2$ if $y_k^1 < y_k^2$, $k = 1, \dots, p$. Furthermore, we shall use $\mathbb{R}_{\geq}^p := \{y \in \mathbb{R}^p : y \geq 0\}$ and analogously $\mathbb{R}_{\leq}^p$ and $\mathbb{R}_{>}^p$.

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The sets X_E and Y_N can be partitioned into two subsets, respectively. An efficient solution $\hat{x} \in X_E$ is called a supported efficient solution if there exists some $\lambda \in \mathbb{R}_{\geq}^p$ such that $\lambda^T C \hat{x} \leq \lambda^T C x$ for all $x \in X$. The set of all supported efficient solutions is denoted X_{sE} , the elements of $X_E \setminus X_{sE} =: X_{nE}$ are called nonsupported efficient solutions. Analogously, we obtain a partition of nondominated points $Y_N = Y_{sN} \cup Y_{nN}$ into supported and nonsupported nondominated points. Note that this definition implies that $Y_{nN} \subset \text{int}((\text{conv } Y_N) + \mathbb{R}_{\geq}^p)$ because there is no hyperplane with gradient in $\mathbb{R}_{\geq}^p$ supporting $\text{conv } Y_N$ at $y \in Y_{nN}$.

Finally, we need to introduce some general (convex analysis) notation for sets in $\mathbb{R}^p$. Let $S \subset \mathbb{R}^p$, then $\text{bd } S$, $\text{int } S$, $\text{cl } S$ denote the boundary, interior, and closure of S , respectively. Furthermore, $\text{conv } S$ denotes the convex hull of S , $S_N := \{s \in S : (s - \mathbb{R}_{\geq}^p) \cap S = \{s\}\}$ is called the nondominated frontier of S and $S^c := \mathbb{R}^p \setminus S$ denotes the complement of S in $\mathbb{R}^p$. S is called $\mathbb{R}_{\geq}^p$ -closed, if the set $S + \mathbb{R}_{\geq}^p := \{s + y : s \in S, y \in \mathbb{R}_{\geq}^p\}$ is closed and $\mathbb{R}_{\geq}^p$ -bounded, if there exists $s^0 \in \mathbb{R}^p$ such that $S \subset s^0 + \mathbb{R}_{\geq}^p$. Finally, S is called $\mathbb{R}_{\geq}^p$ -convex, if $S + \mathbb{R}_{\geq}^p$ is convex.

Lemma 1 will be instrumental to derive results on bound sets.

Lemma 1 (Sawaragi [3]). *Let $S \subset \mathbb{R}^p$.*

1. *If S is an $\mathbb{R}_{\geq}^p$ -closed and $\mathbb{R}_{\geq}^p$ -bounded set then $S \subset S_N + \mathbb{R}_{\geq}^p$ (S is called externally stable).*
2. $S_N = (S + \mathbb{R}_{\geq}^p)_N$.
3. $S_N \subset \text{bd } S$.

1.2. Bounds

In single objective combinatorial optimization lower and upper bounds $l, u \in \mathbb{R}$ with $l \leq \min\{cx : x \in X\} \leq u$ have been essential tools for the solution of $\mathcal{NP}$ -hard problems.

The theory of approximation is based on bounds. They allow the development of algorithms with performance ratios, guaranteeing heuristic solutions to be no more than a certain percentage from optimality. Theoretical investigations lead to results on hardness of approximation. We refer to [4] for an extensive list of results. Bounds are used in implicit enumeration techniques such as branch and bound or dynamic programming for solving combinatorial optimization problems. The performance of the algorithms depends on the quality of the bounds. From this experience, one must assume that bounds might be valuable tools for multiobjective combinatorial optimization, too. However, a different concept of “bounding” is needed to account for the different concept of optimality in multiobjective optimization.

Generally accepted and well known bounds in multiobjective optimization are the ideal point y^I defined by

$$y_k^I := \min_{x \in X} c^k x; \quad k = 1, \dots, p$$

as lower bound and the nadir point y^N defined by

$$y_k^N := \max_{x \in X_E} c^k x; \quad k = 1, \dots, p$$

as upper bound on the value of any efficient point. It is obvious that $y_k^I \leq y_k \leq y_k^N$ for all $y \in Y_N$. The benefits of using ideal and nadir values as bounds are limited due to a number of reasons.

1. y^I is difficult to obtain, if $\min\{c^k x : x \in X\}$ is $\mathcal{NP}$ -hard, as it involves the solution of p $\mathcal{NP}$ -hard problems.
2. y^N is difficult to obtain if $p > 2$ even if the single objective problems $\min\{c^k x : x \in X\}$ are polynomially solvable. Furthermore, heuristics to compute an estimate $\tilde{y}^N$ may underestimate y^N , i.e. $\tilde{y}^N \leq y^N$, rendering the heuristic nadir point useless as an upper bound for Y_N . See [5,6] for discussions on these problems.
3. Last but not least, ideal and nadir values are “far away” from most nondominated points. This is due to the usually large number of nondominated points in Y_N and the conflicting nature of the objectives, whereas y^I and y^N show the combined best and worst values for all objectives over the range of Y_N .

Not much else has been done. Villarreal and Karwan [7] propose the use of sets of lower and upper bound vectors to limit the search space in dynamic programming. Local ideal and nadir points are used for bounding the nondominated

sets of subproblems in nodes of the branch and bound tree, see e.g. [8–10]. Finally, Kaliszewski [11] proposes parametric bounds in interactive problems.

In this paper, following our preliminary investigation in [12,13], we propose a new approach to bounding strategies for (MOCO) problems. Our earlier paper has also been the foundation for further investigation in [14].

We generalize the notion of a bound value to that of a bound set (in parallel to the fundamental transition from unique optimal value to nondominated set), thus incorporating the nature of multiple objectives into the definition of bound sets. Furthermore, we demonstrate that bound sets are not only easier to compute than ideal and nadir points, but also yield better results, i.e. are closer to nondominated points.

The rest of the paper is organized as follows. In Section 2 we give the definitions of lower and upper bound sets and prove some general results. In Section 3 we present techniques for computing lower and upper bound sets for polynomially solvable and $\mathcal{NP}$ -hard combinatorial problems. First, general techniques are given, these are followed by specifications for the assignment problem, the knapsack problem, the set covering problem, the set packing problem and the travelling salesman problem. For these five problems we provide numerical results in Section 4. We use several measures to determine the quality of the bound sets and show typical instances graphically.

2. Definitions and general results

We begin with the definition of lower and upper bound sets. Consider a multiobjective combinatorial optimization problem.

Definition 1. Let $\bar{Y} \subset Y_N$.

1. A lower bound set L for $\bar{Y}$ is an $\mathbb{R}_{\leq}^p$ -closed and $\mathbb{R}_{\leq}^p$ -bounded set $L \subset \mathbb{R}^p$ such that $\bar{Y} \subset L + \mathbb{R}_{\leq}^p$ and $L \subset (L + \mathbb{R}_{\leq}^p)_N$.
2. An upper bound set U for $\bar{Y}$ is an $\mathbb{R}_{\geq}^p$ -closed and $\mathbb{R}_{\geq}^p$ -bounded set $U \subset \mathbb{R}_{\geq}^p$ such that $\bar{Y} \in \text{cl}[(U + \mathbb{R}_{\geq}^p)^c]$ and $U \subset (U + \mathbb{R}_{\geq}^p)_N$.

Note that although the first condition of the definition of a lower bound set seems to be the intuitive formalization of the idea “ L is below $\bar{Y}$ ” this condition alone would still allow for $y \in \bar{Y}, l \in L$ to exist that satisfy $y \in L + \mathbb{R}_{\leq}^p$ but $y < l$, which we want to exclude. The definition of upper bound set is motivated by the idea that any set of feasible solutions $\{x^1, \dots, x^r\} \subset X$ that does not contain x^i, x^j with $Cx^i \leq Cx^j$ or $Cx^j \leq Cx^i$ should define an upper bound set $\{Cx^1, \dots, Cx^r\}$ on Y_N . In particular, $\bar{Y}$ itself is an upper bound set. Since U is $\mathbb{R}_{\geq}^p$ -closed, we must therefore use the closure of the complement of $U + \mathbb{R}_{\geq}^p = \{u + y : u \in U, y \in \mathbb{R}_{\geq}^p\}$ and require that $\bar{Y} \in \text{cl}[(U + \mathbb{R}_{\geq}^p)^c]$.

Using statement 2 in Lemma 1 the second condition for lower (upper) bound sets can be written as $L = \bar{L}_N$ ($U = U_N$). With Definition 1 we can immediately make some (trivial) observations.

1. $\{y^I\}$ is a lower bound set for all $\bar{Y} \subseteq Y_N$.
2. $\{y^N\}$ is an upper bound set for all $\bar{Y} \subseteq Y_N$.

Our first result shows that supported nondominated points define a lower bound set as well as an upper bound set for Y_N .

Proposition 1. 1. $(\text{conv } Y_{sN})_N$ is a lower bound set for Y_N .

2. If L is a lower bound set for Y_{sN} and L is $\mathbb{R}_{\geq}^p$ -convex, then L is a lower bound set for Y_N .

3. Y_{sN} is an upper bound set for Y_N .

Proof. 1. Because X is finite, so is Y_N . Therefore, $\text{conv } Y_{sN}$ is compact, in particular it is $\mathbb{R}_{\leq}^p$ -closed and $\mathbb{R}_{\leq}^p$ -bounded. According to 1 in Lemma 1, $\text{conv } Y_{sN} \subset (\text{conv } Y_{sN})_N + \mathbb{R}_{\leq}^p$. Since $Y_{sN} \subset \text{conv } Y_{sN}$ and $Y_{sN} \subset \text{int}((\text{conv } Y_{sN}) + \mathbb{R}_{\geq}^p)$ we have $Y_N \subset (\text{conv } Y_{sN})_N + \mathbb{R}_{\leq}^p$.

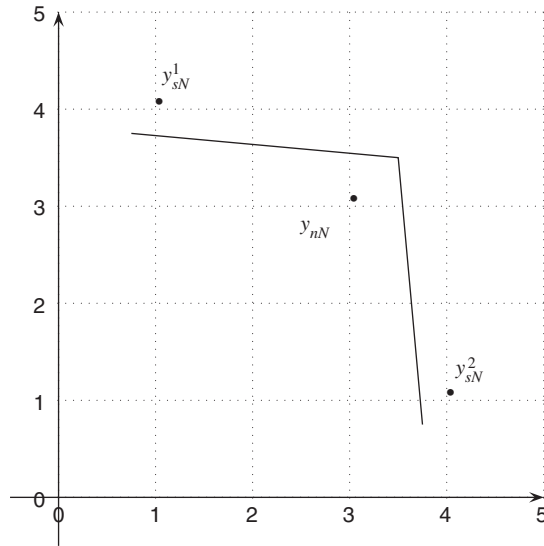


Fig. 1. The line shown is a lower bound set for Y_{sN} and an upper bound set for Y_{nN} but neither lower nor upper bound set for Y_N .

Furthermore, 2 in Lemma 1 yields $(\text{conv } Y_{sN})_N = (\text{conv } Y_{sN} + \mathbb{R}_{\geq}^p)_N$. So it remains to show that $\text{conv } Y_{sN} + \mathbb{R}_{\geq}^p = (\text{conv } Y_{sN})_N + \mathbb{R}_{\geq}^p$. The inclusion “ $\subseteq$ ” follows from external stability of $\text{conv } Y_{sN}$ and the inclusion “ $\supseteq$ ” is trivial.

2. Because $Y_{sN} \subset L + \mathbb{R}_{\geq}^p$ we have $\text{conv } Y_{sN} + \mathbb{R}_{\geq}^p \subset L + \mathbb{R}_{\geq}^p$ due to $\mathbb{R}_{\geq}^p$ -convexity of L . The rest follows from 1.
3. It is enough to observe that no $y \in Y_N$ can be dominated by $y' \in Y_{sN}$. $\square$

Note that the $\mathbb{R}_{\geq}^p$ -convexity in statement 2 is necessary: If L is an arbitrary lower bound set for Y_{sN} then it is not necessarily a lower bound set for Y_N , too, as shown in Fig. 1. Fig. 1 also shows that an analogous result for upper bound sets U of Y_{nN} to be an upper bound set for y_N is not true in general.

Despite the situation shown in Fig. 1 it is true that Y_{nN} itself is an upper bound set for Y_N . To see this it is enough to note that for all $\lambda \in \mathbb{R}_{\geq}^p$ and $y \in Y_{nN}$ we have $\lambda^T y \leq \lambda^T y'$ for all $y' \in y + \mathbb{R}_{\geq}^p$, so that no supported nondominated point can be in $Y_{nN} + \mathbb{R}_{\geq}^p$. By the same reasoning it follows that any upper bound set for Y_{nN} that consists of a finite subset of $Y = CX$ is an upper bound set for Y_N .

Lemma 2. Let L and U be lower and upper bound sets for Y_N and $U \subset Y$. Then $U + \mathbb{R}_{\geq}^p \subset L + \mathbb{R}_{\geq}^p$.

Proof. $U \subset Y \subset Y_N + \mathbb{R}_{\geq}^p \subset L + \mathbb{R}_{\geq}^p$. $\square$

In single objective optimization, bounding procedures can simply be combined by choosing the one that gives the best result in any given situation. In the multicriteria case, some more care is needed. Let L_1, L_2 and U_1, U_2 be two lower, respectively, upper bound sets for $\bar{Y}$. We say L_1 dominates L_2 if $L_1 \subset L_2 + \mathbb{R}_{\geq}^p$ and $L_1 + \mathbb{R}_{\geq}^p \neq L_2 + \mathbb{R}_{\geq}^p$. U_1 dominates U_2 if $U_2 \subset U_1 + \mathbb{R}_{\geq}^p$ and $U_2 + \mathbb{R}_{\geq}^p \neq U_1 + \mathbb{R}_{\geq}^p$.

Proposition 2. 1. If L_1, L_2 are lower bound sets for $\bar{Y}$ and $L_1 + \mathbb{R}_{\geq}^p \neq L_2 + \mathbb{R}_{\geq}^p$ then

$$L^* := [(L_1 + \mathbb{R}_{\geq}^p) \cap (L_2 + \mathbb{R}_{\geq}^p)]_N$$

is a lower bound set dominating L_1 and L_2 .

2. If U_1, U_2 are upper bound sets for $\bar{Y}$, $U_1 + \mathbb{R}_{\geq}^p \neq U_2 + \mathbb{R}_{\geq}^p$ then

$$U^* := [(U_1 + \mathbb{R}_{\geq}^p) \cup (U_2 + \mathbb{R}_{\geq}^p)]_N$$

is an upper bound set dominating U_1 and U_2 .

Proof. 1. Since $\bar{Y} \subset L_1 + \mathbb{R}_{\geq}^p$ and $\bar{Y} \subset L_2 + \mathbb{R}_{\geq}^p$, $\bar{Y}$ is contained in their intersection. Because L_1 and L_2 are both $\mathbb{R}_{\geq}^p$ -closed and $\mathbb{R}_{\geq}^p$ -bounded, so is their intersection and therefore L^* . That $L^* = L_N^*$ follows by definition of L^* . The inclusion of L^* in $L_1 + \mathbb{R}_{\geq}^p$ and $L_2 + \mathbb{R}_{\geq}^p$ is trivial. The final condition for L^* dominating L_1 and L_2 follows from $L_1 + \mathbb{R}_{\geq}^p \neq L_2 + \mathbb{R}_{\geq}^p$.

2. The proof is similar to the proof of 1. $\square$

Naturally, we will be interested in bound sets that are “as close as possible” to the set $\bar{Y}$ in question. To that end, we say that a lower bound set L (upper bound set U) is *perfect* if

$$\bar{Y} \subset L \quad (\bar{Y} \subset U).$$

Proposition 3. A lower bound set L for $\bar{Y}$ is perfect if and only if $\bar{Y} \subset (L + \mathbb{R}_{\geq}^p)_N$. If U is a perfect upper bound set for $\bar{Y}$ then $\bar{Y} \subset (U + \mathbb{R}_{\geq}^p)_N$. If, in addition, $U \subset Y$ then $\bar{Y} \subset (U + \mathbb{R}_{\geq}^p)_N$ is sufficient for U to be perfect.

Proof. The statement for lower bound sets follows because $L = L_N = (L + \mathbb{R}_{\geq}^p)_N$ by the definition of lower bound set. The statement for upper bound sets follows because $U = U_N = (U + \mathbb{R}_{\geq}^p)_N$ by definition of an upper bound set. It is easy to see that the condition is not sufficient for upper bound sets in general. Assume that $U \subset Y$. Then $\bar{Y} \subset (U + \mathbb{R}_{\geq}^p)_N \subset \text{bd}(U + \mathbb{R}_{\geq}^p) = U + \text{bd } \mathbb{R}_{\geq}^p$. Because $\bar{Y} \subset Y_N$ and $U \subset Y$ this implies $\bar{Y} \subset U$. $\square$

It is obvious that $(\text{conv } Y_{sN})_N$ is a perfect lower bound set for Y_{sN} . Next we consider general $\mathbb{R}_{\geq}^p$ -convex lower bound sets for Y_N .

Proposition 4. An $\mathbb{R}_{\geq}^p$ -convex lower bound set L for Y_N is perfect only if $Y_{nN} = \emptyset$.

Proof. $Y_N \subset (L + \mathbb{R}_{\geq}^p)_N \subset \text{bd}(L + \mathbb{R}_{\geq}^p) = \text{bd}(\text{conv } L + \mathbb{R}_{\geq}^p)$ by Proposition 3, Lemma 1 and $\mathbb{R}_{\geq}^p$ -convexity of L . On the other hand we know from the definition of Y_{nN} and Lemma 2 that $Y_{nN} \subset \text{int}(L + \mathbb{R}_{\geq}^p)$. $\square$

With Propositions 1 and 4 $(\text{conv } Y_{sN})_N$ is the “best” $\mathbb{R}_{\geq}^p$ -convex bound set for Y_N . As a consequence, to find non-supported nondominated points nonconvex bound sets will be needed. This has implications for, e.g., the development of branch-and-bound algorithms for multiobjective combinatorial optimization problems.

Finally, we show the biobjective analogon of the fact “lower bound equals upper bound implies optimality”.

Proposition 5. Let $p = 2$ and let $L, U \subset Y$ be a lower and an upper bound set for $\bar{Y}$, respectively. If $\text{int}[(L + \mathbb{R}_{\geq}^2) \cap \text{cl}((U + \mathbb{R}_{\geq}^2)^c)] = \emptyset$ then both bound sets are perfect.

Proof. L and U are lower and upper bound sets for $\bar{Y}$ therefore $\bar{Y} \subset (L + \mathbb{R}_{\geq}^2) \cap \text{cl}((U + \mathbb{R}_{\geq}^2)^c)$. Thus, we must have $\bar{Y} \subset \text{bd}(L + \mathbb{R}_{\geq}^2)$ and $\bar{Y} \subset \text{bd}(U + \mathbb{R}_{\geq}^2)^c$.

We first show that $\bar{Y} \subset L$. Let $y \in \bar{Y}$. Then there are $l \in L$ and $d \in \mathbb{R}_{\geq}^2$ such that $y = l + d$. If $d > 0$ we get $y \in \text{int}(L + \mathbb{R}_{\geq}^2)$. On the other hand, $y \in (U + \mathbb{R}_{\geq}^2)^c$ implies that $y - d \in \text{int}(U + \mathbb{R}_{\geq}^2)^c$. Thus $z = l + 0.5d = y - 0.5d \in \text{int}(L + \mathbb{R}_{\geq}^2) \cap \text{int}(U + \mathbb{R}_{\geq}^2)^c$, contradicting the assumption.

Therefore, $d \in \mathbb{R}_{\leq}^2 \setminus \mathbb{R}_{<}^2$. Then there is $u \in \text{bd}(U + \mathbb{R}_{\leq}^2)$ such that $u_2 = y_2$ or $u_1 = y_1$. Let $d' = u - y$, then $d' \in \text{bd}(\mathbb{R}_{\leq}^2 \setminus \{0\})$. Thus $l + 0.5d + 0.5d' = u - 0.5d - 0.5d' \in \text{int}(L + \mathbb{R}_{\leq}^2) \cap \text{int}(U + \mathbb{R}_{\leq}^2)^c$ contradicting the assumption.

Hence, $d = 0$ and $y \in L$.

To see that $\bar{Y} \subset U$ note that $\bar{Y} \subset \text{bd}(U + \mathbb{R}_{\leq}^2)^c = \text{bd}(U + \mathbb{R}_{\leq}^2) = U + \text{bd}(\mathbb{R}_{\leq}^2)$ because $U \subset \text{bd}(U + \mathbb{R}_{\leq}^2)$ by definition of a bound set and Lemma 1. But $y = u + d$ for $y \in \bar{Y}$, $u \in U$, and $d \in \text{bd}(\mathbb{R}_{\leq}^2)$ implies $d = 0$ since $U \subset Y$ and $\bar{Y} \subset Y_N$. $\square$

Note that the assumption made in Proposition 5 is very strong. Even in the case $L = (\text{conv } Y_{sN})_N$, $U = Y_{sN}$ and $Y_{nN} = \emptyset$ it is not satisfied because $L + \mathbb{R}_{\leq}^p$ is a convex set, but $\text{cl}((U + \mathbb{R}_{\leq}^2)^c)$ is not (the “upper right boundary” of this set is a step function), so that the interior of their intersection is nonempty. This illustrates once more, that a restriction to convex (lower) bound sets is limiting.

3. Constructing bound sets for MOCO

In this section we present techniques for constructing lower and upper bound sets for problems with two objectives. The general idea to construct bound sets is by using the weighted sum scalarized problem

$$\min_{x \in X} \lambda c^1 x + (1 - \lambda) c^2 x. \quad (P_\lambda)$$

We distinguish the cases where the single objective problems $\min\{c^k x : x \in X\}$ are polynomially solvable, respectively, $\mathcal{NP}$ -hard. Note that the distinction applies to the single objective case. The biobjective case is usually $\mathcal{NP}$ -hard even for such problems as assignment, shortest path, spanning tree etc., see [15].

For polynomially solvable problems this idea yields both a lower and an upper bound set for Y_N . In the $\mathcal{NP}$ -hard case, we employ relaxations to obtain lower bounds for (P_λ) and heuristics for upper bounds.

3.1. Bound sets for polynomially solvable single objective problems

In this case, we can solve (P_λ) in polynomial time, therefore we can find Y_{sN} exactly, which provides $L = (\text{conv } Y_{sN})_N$ as lower bound set for y_N as well as an upper bound set $U = Y_{sN}$.

L is called the convex hull lower bound set. We shall call U the local nadir upper bound set, because it is defined by the nadir point given by two adjacent supported nondominated points. Let $Y_{sN} = \{y^1, \dots, y^r\}$ such that $y_1^1 < y_1^2 < \dots < y_1^r$ (and therefore $y_2^1 > y_2^2 > \dots > y_2^r$). Then $(L + \mathbb{R}_{\leq}^p) \setminus (U + \mathbb{R}_{\leq}^p)$ consists of triangles with corner points y^i, y^{i+1} and $(y_1^{i+1}, y_2^i)^T$, $i = 1, \dots, r - 1$. The point $(y_1^{i+1}, y_2^i)^T$ is a local nadir point defined by two adjacent supported nondominated points y^i and y^{i+1} .

It is worthwhile noting that although (P_λ) can be solved in polynomial time, the complexity of finding Y_{sN} is $O(|Y_{sN}|p(n, m))$, where $p(n, m)$ is a polynomial in the problem size. The cardinality of Y_{sN} can be exponential in problem size, see, e.g. [16].

The procedure for finding L can now be chosen as the first phase of the two phase procedure described in [17,18]. It is listed below as Algorithm 1.

Algorithm 1

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firstPhase: procedure ()
  --| Compute  $x^{(1)}$  and  $x^{(2)}$ , optimal solutions of
  --|  $(P_\lambda)$  for  $\lambda = 1$  and  $\lambda = 0$ 
   $x^{(1)} \leftarrow \text{Solve } P_\lambda(1 \downarrow)$ 
   $x^{(2)} \leftarrow \text{Solve } P_\lambda(0 \downarrow)$ 
   $S \leftarrow \{x^{(1)}, x^{(2)}\}$ 
  --| Compute all supported solutions between  $x^{(1)}$  and  $x^{(2)}$ .
  --| Update  $S$  with all new nondominated solutions generated.
  solveRecursion( $x^{(1)} \downarrow, x^{(2)} \downarrow, S \uparrow$ )
end firstPhase

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solveRecursion: procedure ( $x^{(A)} \downarrow, x^{(B)} \downarrow, S \uparrow$ )
-| Compute optimal solutions  $x^{(C)}$  of  $(P_\lambda)$ :
-|  $\min\{\lambda_1 z^1(x) + \lambda_2 z^2(x) \mid x \in X\}$ , where
-|  $\lambda_1 = z^2(x^{(A)}) - z^2(x^{(B)})$ , and  $\lambda_2 = z^1(x^{(B)}) - z^1(x^{(A)})$ .
 $x^{(C)} \leftarrow \text{Solve } P_\lambda(\lambda \downarrow)$ 
if  $x^{(C)} \notin S$  then
   $S \leftarrow (S \cup \{x^{(C)}\})_N$ 
  solveRecursion( $x^{(A)} \downarrow, x^{(C)} \downarrow, S \uparrow$ )
  solveRecursion( $x^{(C)} \downarrow, x^{(B)} \downarrow, S \uparrow$ )
end if
end solveRecursion

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While the same lower bound set is found if either one or all optimal solutions of (P_λ) are found, the upper bound set will be better (in the sense of domination, see before Proposition 2) if all optimal solutions are considered.

We have used this procedure for the biobjective assignment problem, see Section 4.1.

When the single objective problems $\{\min c^k x : x \in X\}$ are $\mathcal{NP}$ -hard, we cannot use this algorithm directly, but have to resort to relaxations and heuristics for (P_λ) instead. The necessary modifications are described in Section 3.2.

3.2. Bound sets for $\mathcal{NP}$ -hard single objective problems

A much more interesting problem is the computation of bound sets for problems which are $\mathcal{NP}$ -hard in the single objective case. We address lower bound sets first. Then, we cannot compute $(\text{conv } Y_{sN})_N$ because (P_λ) is itself an $\mathcal{NP}$ -hard problem. Instead of solving (P_λ) exactly, we compute a lower bound by solving a relaxed problem, which yields some $x^\lambda \in \mathbb{R}^p$, not necessarily contained in X , and compute $c^k x^\lambda$, $k = 1, 2$.

Suppose that we have a relaxation of (MOCO), that is a multiobjective optimization problem

$$\min\{\tilde{C}x : x \in \tilde{X}\} \quad (1)$$

such that the feasible set X of (MOCO) is contained in $\tilde{X}$ and for the nondominated sets we have $Y_N \subset \text{conv } \tilde{Y}_N + \mathbb{R}_{\geq}^p$, where $\tilde{Y}_N$ is the nondominated set of (1). Let $(\tilde{P}_\lambda)$ denote the problem $\min\{\lambda^T \tilde{C}x : x \in \tilde{X}\}$.

Lemma 3. Consider a relaxation (1) of (MOCO). For all $\lambda \in \mathbb{R}_{\geq}^p$ $(\tilde{P}_\lambda)$ is a relaxation of (P_λ) .

Proof. The feasible set of (P_λ) , namely X , is contained in the feasible set of $(\tilde{P}_\lambda)$ by definition. Consider $y \in Y_N$. Then there are $y' \in \text{conv } \tilde{Y}_N$ and $d \in \mathbb{R}_{\geq}^p$ such that $y = y' + d$. Therefore, for all $\lambda \in \mathbb{R}_{\geq}^p$ we have $\lambda^T y = \lambda^T y' + \lambda^T d$. Since $\lambda, d \in \mathbb{R}_{\geq}^p$ we get $\lambda^T y' \leq \lambda^T y$. If y' is not in $\tilde{C} \tilde{X}$ there is some $x \in \tilde{X}_{sE}$ with $\lambda^T y' = \lambda^T \tilde{C}x$, so $\min_{x \in \tilde{X}} \lambda^T \tilde{C}x \leq \min_{x \in X} \lambda^T Cx$. $\square$

The weighted sum scalarized problem $(\tilde{P}_\lambda)$ of (1) is thus a relaxation of (P_λ) . In the biobjective case we can apply Algorithm 1 to find the supported efficient solutions of (1), which can be used to define a lower bound set $(\text{conv } \tilde{Y}_{sN})_N$.

For practical purposes we shall assume that the feasible set $\tilde{X}$ of the relaxation (1) is polyhedral or finite, so that $(\text{conv } \tilde{Y}_N)$ has a finite number of extreme points and Algorithm 1 is guaranteed to terminate. At termination we have a lower bound set given by

$$L = (\text{conv } \tilde{Y}_{sN})_N, \quad (2)$$

which in the biobjective case is obviously a monotone decreasing, convex (piecewise linear) curve. That L is a lower bound set for Y_N is true because $Y_{sN} \subset L + \mathbb{R}_{\geq}^2$ and by Proposition 1. We call L a convex hull lower bound set.

Sometimes it might be possible to improve a lower bound for (P_λ) , but impossible to decompose that bound into two components unless it is given by $x \in \tilde{X} \cap X$. An example of this behaviour is given in Section 4.3, where we use Lagrangian relaxation and subgradient optimization for (P_λ) in the case of the TSP.

In this situation a few modifications of Algorithm 1 are necessary. In the initialization step solutions x^1, x^2 of the relaxation $(\tilde{P}_{(1,0)})$ and $(\tilde{P}_{(0,1)})$ are computed and $S_1 := S = \{Cx^1, Cx^2\}$. We require a second set S_2 to store those points that are not on the convex hull of S_1 . Assume that $S_1 = \{Cx^1, \dots, Cx^r\}$ at some stage of the algorithm and the relaxation of (P_λ) yields x^λ .

1. If Cx^λ is dominated by some $y^j \in S_1 \cup S_2$, y^j is discarded (because $\lambda^T Cx^\lambda \geq \lambda^T y^j$ for all $\lambda \in \mathbb{R}_{\geq}^p$).
2. If Cx^λ dominates some $y^j \in S_1 \cup S_2$, Cx^λ is discarded.
3. If Cx^λ is not dominated and does not dominate any element of $S_1 \cup S_2$ then if $Cx^\lambda \in \text{conv } S_1$ add it to S_2 , otherwise add it to S_1 .

The algorithm uses only S_1 to compute new λ values in the recursive procedure. If S_2 is nonempty, the lower bound set can be improved in the following way:

$$L' := \left((L + \mathbb{R}_{\geq}^2) \setminus \bigcup_{y \in S_2} (y - \mathbb{R}_{\geq}^2) \right)_N. \quad (3)$$

In L' we simply use the fact that no point of Y_N can be contained in $y - \mathbb{R}_{\geq}^2$, $y \in S_2$, as it would be better than y for any (P_λ) . In fact, L' is also a lower bound set for Y_N . Note that L' is not $\mathbb{R}_{\geq}^2$ -convex.

Another strategy for computing a lower bound set is as follows. Choose a finite set of values $\lambda_1^i \in [0, 1]$, $i = 1, \dots, r$ such that $0 = \lambda_1^1 < \dots < \lambda_1^r = 1$. Let $\lambda^i = (\lambda_1^i, 1 - \lambda_1^i)$ and let x^i be a solution of some relaxation of (P_{λ^i}) . Then

$$Y^i := \{y \in \mathbb{R}^2 : \lambda^i{}^T y = \lambda^i{}^T Cx^i\}, \quad i = 2, \dots, r-1$$

as well as $Y^1 := \{y \in \mathbb{R}^2 : y_2 = c^2 x^1\}$ and $Y^r := \{y : y_1 = c^1 x^r\}$ are lower bound sets for Y_N by definition of x^i . Therefore, according to Proposition 2,

$$L'' := \left(\bigcap_{i=1}^r Y^i + \mathbb{R}_{\geq}^2 \right)_N \quad (4)$$

is a lower bound set for Y_N . In fact, L'' is the lower envelope of the hyperplanes Y^i , $i = 1, \dots, r$. We call it the hyperplane bound set. It is $\mathbb{R}_{\geq}^2$ -convex. This technique can also be applied for $p > 2$.

As for upper bound sets, we observe that in contrast to the case of polynomially solvable problems, we cannot use the triangles defined by adjacent extreme points of L to derive an upper bound set. This is due to L consisting of infeasible points, and is clearly seen on the illustrations in Section 4.

However, we can employ any heuristic to find feasible solutions of (P_λ) and again modify Algorithm 1 to deal with dominated points and nondominated points not on the convex hull of points computed so far. Again, we use two sets S_1 and S_2 , where S_1 contains points used to compute λ values and S_2 contains points inside $\text{conv } S_1$. Let x^λ be a heuristic feasible solution of (P_λ) .

1. If Cx^λ is dominated by some $y^j \in S_1 \cup S_2$, Cx^λ is discarded.
2. If Cx^λ dominates some $y^j \in S_1 \cup S_2$, y^j is discarded.
3. If Cx^λ is not dominated and does not dominate any element of $S_1 \cup S_2$ then if $Cx^\lambda \in \text{conv } S_1$ add it to S_2 , otherwise add it to S_1 .

Let us now look at the quality of the bound sets obtained by using the methods mentioned above. We assume that the algorithm to solve $(\tilde{P}_\lambda)$ has a performance ratio $1 < \rho_L$, i.e. for all λ it is possible to find $x_\lambda \in \tilde{X}$ such that $\lambda^T \tilde{C}x_\lambda \leq \min_{x \in X} \lambda^T Cx \leq \rho_L \lambda^T \tilde{C}x_\lambda$.

Lemma 4. Let L be a convex hull lower bound set. For $l \in L$ let

$$A(l) := \left\{ \lambda \in \mathbb{R}_{\geq}^2 : \lambda^T l \leq \min_{x \in \tilde{X}} \lambda^T \tilde{C}x \right\}.$$

Assume that the algorithm for $(\tilde{P}_\lambda)$ has performance ratio ρ_L . Then for all $\lambda \in A(l)$ it holds

$$\lambda^T l \leq \min_{x \in X} \lambda^T Cx \leq \rho_L \lambda^T l.$$

Proof. Let $\lambda \in A(l)$. Thus $\lambda^T l \leq \min_{x \in \tilde{X}} \lambda^T Cx$. Let $\tilde{x}$ be an optimal solution of $(\tilde{P}_\lambda)$. Then

$$\lambda^T l = \lambda^T \tilde{C} \tilde{x} \leq \min_{x \in X} \lambda^T Cx \leq \rho_L \lambda^T \tilde{C} \tilde{x} = \rho_L \lambda^T l. \quad \square$$

If L is a convex hull lower bound set, we can employ the fact that $\mathbb{R}_{\geq}^2 = \cup_{l \in L} A(l)$ to prove an approximation for all supported efficient solutions.

Proposition 6. Under the assumptions of Lemma 4 it holds that for all $y \in Y_{sN}$ there is $\lambda \in \mathbb{R}_{\geq}^2$ and $l \in L$ such that $\lambda^T l \leq \min_{x \in X} \lambda^T Cx \leq \rho_L \lambda^T l$.

Proof. Let $y \in Y_{sN}$ be a supported nondominated point. Then there is some $\lambda \in \mathbb{R}_{\geq}^2$ such that $\lambda^T y \leq \min_{x \in X} \lambda^T Cx$. For this λ there is $l \in L$ such that $\lambda \in A(l)$. By Lemma 4 $\lambda^T l \leq \min_{x \in X} \lambda^T Cx \leq \rho_L \lambda^T l$. $\square$

Proposition 6 implies that the quality of single objective lower bounds transfers to the convex hull lower bound sets L . We can therefore speak of lower bound sets for supported efficient solutions with performance ratio ρ_L . Note, however, that the performance ratio is not necessarily valid for Y_N , as we do not have any information about the distance of unsupported nondominated points from the bound set. Also, although Lemma 4 holds for other bound sets, e.g. the hyperplane lower bound set, Proposition 6 does not necessarily hold. For the hyperplane bound set there might be $\lambda \in \mathbb{R}_{\geq}^p$ such that L does not contain l with $\lambda \in A(l)$.

An analogous result can be obtained for upper bound sets as follows. Let y_λ^* denote the optimal value of (P_λ) . A heuristic has performance ratio $1 < \rho_U$ if it finds $x \in X$ such that $\lambda^T Cx \leq \rho_U y_\lambda^*$. Then for an upper bound set $U \subset Y$ and $u \in U$ we may define

$$A(u) := \{\lambda \in \mathbb{R}_{\geq}^p : \text{The heuristic for } (P_\lambda) \text{ yields } u\}.$$

Then it is obvious that for all $\lambda \in A(u)$

$$\min_{x \in X} \lambda^T Cx \leq \lambda^T u \leq \rho_U \min_{x \in X} \lambda^T Cx.$$

Moreover, the following proposition holds.

Proposition 7. Let U be an upper bound set computed by Algorithm 1 using a heuristic for (P_λ) with performance ratio ρ_U . Then for every $y \in Y_{sN}$ there are $\lambda \in \mathbb{R}_{\geq}^2$ and $u \in U$ such that $\lambda^T u \leq \rho_U \min_{x \in X} \lambda^T Cx$.

Proof. Let $S_1 = \{u^1, \dots, u^r\}$ with $u_1^i \leq u_1^{i+1}$, $i = 1, \dots, r-1$ denote the set S_1 at termination of the algorithm. Let λ^i be the λ value for which u^i has been found and denote $\lambda^{i,i+1}$ the normal to the line connecting u^i and u^{i+1} .

The algorithm has applied the heuristic to $(P_{\lambda^{i,i+1}})$. Let u be the point found. Then 2 cases are possible.

1. $\lambda^T u \geq \lambda^T u^i = \lambda^T u^{i+1}$. The performance ratio of the heuristic implies that $\min_{x \in X} \lambda^T Cx \leq \lambda^T u \leq \rho_U \min_{x \in X} \lambda^T Cx$. In this inequality u can be replaced by u^i or u^{i+1} . In addition, u^i will satisfy the performance ratio for all λ with $\lambda_1^{i+1} \leq \lambda_1 \leq \lambda_1^{i,i+1}$ and u^{i+1} will do the same for all λ with $\lambda_1^{i,i+1} \leq \lambda_1 \leq \lambda_1^i$.
2. $\lambda^T u < \lambda^T u^i = \lambda^T u^{i+1}$. Then u would have been added to S_1 , but later removed by addition of other points. Since the algorithm cannot have found u' dominating u^i or u^{i+1} , the new points added would have led either to removal of u^i or u^{i+1} from S_1 or contradict the fact that u^i and u^{i+1} are consecutive points in S_1 .

We have therefore shown that for all $\lambda \in \mathbb{R}_{\geq}^2$ with $\lambda_1^{i+1} \leq \lambda_1 \leq \lambda_1^i$ the algorithm has found a feasible solution within ρ_U of the optimal solution of (P_λ) .

Because for any $y \in Y_{sN}$ there is some λ so that $\lambda^T y \leq \min_{x \in X} \lambda^T Cx$, and that λ satisfies $\lambda_1^{i+1} \leq \lambda_1 \leq \lambda_1^i$ for some $i \in \{1, \dots, r-1\}$ the proof is complete. $\square$

4. Numerical results

We have performed numerical tests on all problem types for which instances were available in the MOCO numerical instances library [19] at the time of writing. In this section we present these results, mainly to illustrate that the use of bound sets can be beneficial in multiobjective combinatorial optimization. To measure the quality of lower and upper bound sets, we used a number of measures.

Let L and U be an upper and lower bound set for an instance of a biobjective combinatorial optimization problem. Let $y^{\max} := (\max_{y \in L \cup U} y_1, \max_{y \in L \cup U} y_2)$ and $y^{\min} := (\min_{y \in L \cup U} y_1, \min_{y \in L \cup U} y_2)$, let U' denote $\text{bd}[(U + \mathbb{R}_{\geq}^p) \cap (y^{\max} - \mathbb{R}_{\geq}^p)]$. We define the distance between L and U as $d(L, U) := \max_{u \in U'}, \min_{l \in L} \|u - l\|_2$. A distance based quality measure is

$$\mu_1 = \frac{d(L, U)}{\|y^{\max} - y^{\min}\|_2}.$$

$\|\cdot\|_2$ denotes the Euclidean norm. On Figs. 2–3 two circles with a line inbetween indicate the points that define $d(L, U)$.

We also used the area based measures introduced in [14]. Let A_L denote the area of the set $(L + \mathbb{R}_{\geq}^p) \cap (y^{\max} - \mathbb{R}_{\geq}^p)$, A_U is defined analogously. Furthermore, let A_L^d and A_U^d denote the area of the nondominated triangles with respect to the lower and upper bound sets, respectively. Then

$$\mu_2 := \frac{A_L - A_U}{A_L},$$

$$\mu_3 := \frac{A_U^d}{A_L - A_U},$$

$$\mu_4 := \frac{A_L^d}{A_L - A_U},$$

$$\mu_5 := \frac{\mu_2}{\mu_3}.$$

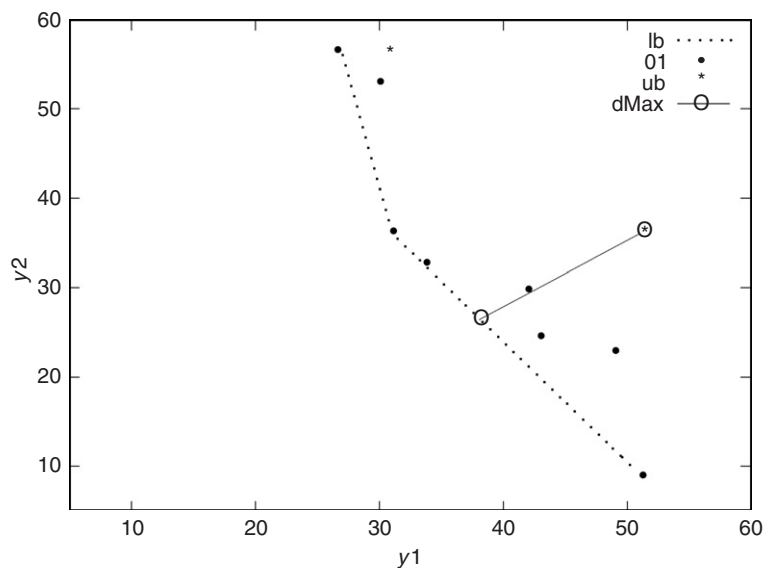


Fig. 2. Assignment problem instance 2AP05-1.

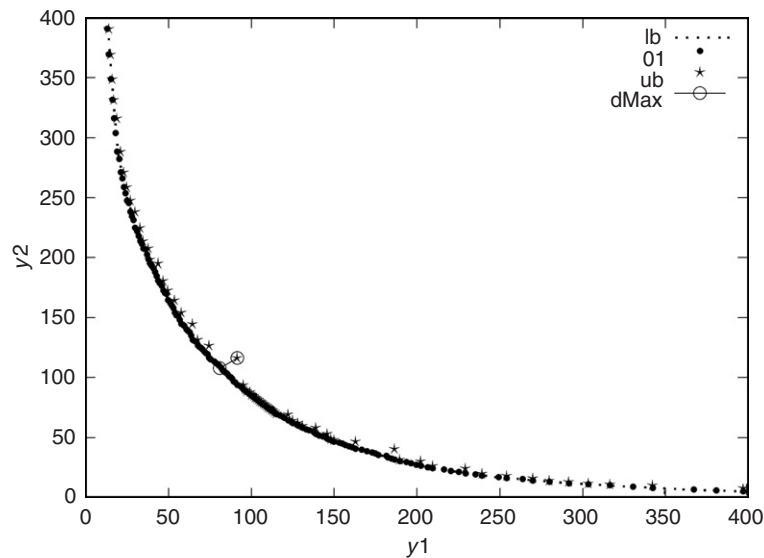


Fig. 3. Assignment problem instance 2AP50-1.

If L and U are “good” lower and upper bound sets, we expect that (qualitatively speaking)

- μ_1 is small. The worst distance between a point on U and L is small with respect to the range of objective values.
- μ_2 is small. Only a small fraction of the area dominated by L is not dominated by U , i.e. the area where additional nondominated points can be found is relatively small.
- μ_3 is big. Most of the area between L and U is covered by nondominated triangles, where additional nondominated points may exist.
- $\mu_4 \geq 1$, which is the case if $U \subset A_L^4$.
- μ_5 is very small (μ_3 big, μ_2 small). Note that μ_3 can be zero only if U consists of a single point. We expect this does not happen in practice.

While there is some literature on evaluating the performance of multiobjective heuristics (see [20,21] and references in the latter) or the quality of a representation of the efficient set [22] this is not quite appropriate here. The reason for this is that the measures proposed in these references do not consider comparison between lower and upper bound sets. These measures usually allow comparisons of the quality of different sets of feasible points as approximations of the true (unknown) nondominated set. Interpreting them as a measure of the difference between a lower and upper bound set can be misleading, since a lower bound set typically consists of points that do not even correspond to feasible solutions of the problem. Comparison of approximations of the nondominated set, on the other hand, means comparing feasible points, i.e., comparison of several upper bound sets. Moreover, the purpose here is not to evaluate heuristic methods, but just to see if it is possible to say that lower and upper bound sets provide a good indication about the location of (the unknown) nondominated points. Thus, we need measures that play a role analogous to the IP gap in single objective integer programming.

In the following sections we present results on the computation of bound sets for five problem types, the assignment problem, the knapsack problem, the travelling salesperson problem, and the set covering and set packing problems. For each, we chose 10 instances of varying characteristics from [19]. Details about the instances can be found there and in referenced papers. The evaluation of the measures μ_1 to μ_5 can be found in Tables 1–5. To illustrate a number of aspects we provide two figures for each problem (Figs. 2–11). For all but the TSP we have also computed the true set of efficient solutions using scalarizations with constraints on the objective values and CPLEX as integer linear programming solver. In the key of the figures lb stands for lower bound, ub for upper bound, 01 for the exact nondominated set, and dMax for $d(L, U)$. Further comments on the figures are made in the appropriate sections below.

Table 1
Results for the assignment problem

Instance	μ_1	μ_2	μ_3	μ_4	μ_5
2ap05-1	0.30454	1.00000	0.28153	0.02075	3.55205
2ap10-1	0.09176	0.30540	0.27915	0.03726	1.09405
2ap15-1	0.05556	0.14608	0.28630	0.03362	0.51023
2ap20-1	0.06788	0.13631	0.31671	0.03166	0.43041
2ap25-1	0.03705	0.06181	0.26045	0.05237	0.23731
2ap30-1	0.03670	0.07045	0.28369	0.04668	0.24833
2ap35-1	0.05551	0.05042	0.24551	0.05319	0.20538
2ap40-1	0.02273	0.03182	0.26816	0.08479	0.11865
2ap45-1	0.01677	0.02565	0.25014	0.07946	0.10253
2ap50-1	0.02490	0.02256	0.27447	0.08743	0.08218

Table 2
Results for the knapsack problem

Instance	μ_1	μ_2	μ_3	μ_4	μ_5
2kp50-1a	0.02769	0.02533	1.31981	2.00849	0.01919
2kp100-1a	0.00959	0.02738	0.52984	1.14146	0.05167
2kp150-1a	0.00842	0.01888	0.54503	1.09636	0.03464
2kp200-1a	0.00507	0.00961	0.75618	2.10240	0.01270
2kp250-1a	0.00544	0.01282	0.38211	0.86068	0.03354
2kp300-1a	0.00590	0.00563	0.82000	2.01198	0.00687
2kp350-1a	0.00392	0.00875	0.45516	1.44900	0.01922
2kp400-1a	0.00406	0.00961	0.35962	1.01855	0.02672
2kp450-1a	0.00325	0.00719	0.35697	1.10325	0.02014
2kp500-1a	0.00373	0.00658	0.36828	1.23589	0.01787

Table 3
Results for the travelling salesperson problem

Instance	μ_1	μ_2	μ_3	μ_4	μ_5
kroab50	0.06195	0.13951	0.08350	0.10747	1.67086
kroac50	0.06741	0.08270	0.14396	0.08823	0.57443
kroad50	0.06741	0.07533	0.10717	0.07662	0.70293
kroae50	0.05528	0.11693	0.05067	0.05633	2.30759
krobc50	0.04396	0.07671	0.09745	0.04098	0.78719
krobd50	0.04926	0.10886	0.07351	0.09141	1.48075
krobe50	0.04389	0.06270	0.15034	0.04208	0.41707
krocd50	0.04828	0.07288	0.16291	0.06589	0.44736
kroce50	0.05147	0.08384	0.13670	0.05901	0.61332
krode50	0.03853	0.06000	0.13842	0.05191	0.43349

4.1. The assignment problem

The assignment problem (5) is the only polynomially solvable problem considered.

$$\begin{aligned}
 \min \quad & \left(\sum_{i=1}^n \sum_{j=1}^n c_{ij}^1 x_{ij}, \sum_{i=1}^n \sum_{j=1}^n c_{ij}^2 x_{ij} \right) \\
 \text{s.t.} \quad & \sum_{i=1}^n x_{ij} = 1, \quad j = 1, \dots, n, \\
 & \sum_{j=1}^n x_{ij} = 1, \quad i = 1, \dots, n, \\
 & x_{ij} \in \{0, 1\}, \quad i, j = 1, \dots, n;
 \end{aligned} \tag{5}$$

Table 4
Results for the set covering problem

Instance	μ_1	μ_2	μ_3	μ_4	μ_5
2scp41a	0.09493	0.13343	0.13265	0.00180	1.00594
2scp41c	0.07556	0.24453	0.31902	0.00192	0.76648
2scp43a	0.05198	0.10636	0.24922	0.00471	0.42675
2scp43c	0.14846	0.31063	0.24852	0.00465	1.24993
2scp61a	0.03580	0.06542	0.10732	0.00060	0.60955
2scp61c	0.04755	0.37177	0.34416	0.00055	1.08023
2scp62a	0.05658	0.07360	0.20121	0.00168	0.36577
2scp81a	0.03767	0.04516	0.22458	0.00072	0.20108
2scp81c	0.01466	0.03586	0.07765	0.00213	0.46184
2scp82a	0.04537	0.04939	0.16099	0.00138	0.30677

Table 5
Results for the set packing problem

Instance	μ_1	μ_2	μ_3	μ_4	μ_5
2mis100_300a	0.41919	0.86352	0.01334	0.00000	64.70758
2mis100_500a	0.61491	0.97660	0.00455	0.00000	214.71121
2mis200_600a	0.49006	0.87619	0.00518	0.00004	169.19485
2mis200_1000a	0.60604	0.96639	0.00627	0.00000	154.13699
2mis201_600a	0.49692	0.89111	0.00927	0.00004	96.12836
2spp100_300a	0.38824	0.81578	0.04506	0.00112	18.10597
2spp100_500a	0.56869	0.95024	0.00825	0.00043	115.11168
2spp200_1000a	0.56402	0.94733	0.01647	0.00066	57.52483
2spp201_600a	0.45252	0.76525	0.01529	0.00213	50.03536
2spp203_600a	0.69410	0.99995	0.00244	0.00114	409.48718

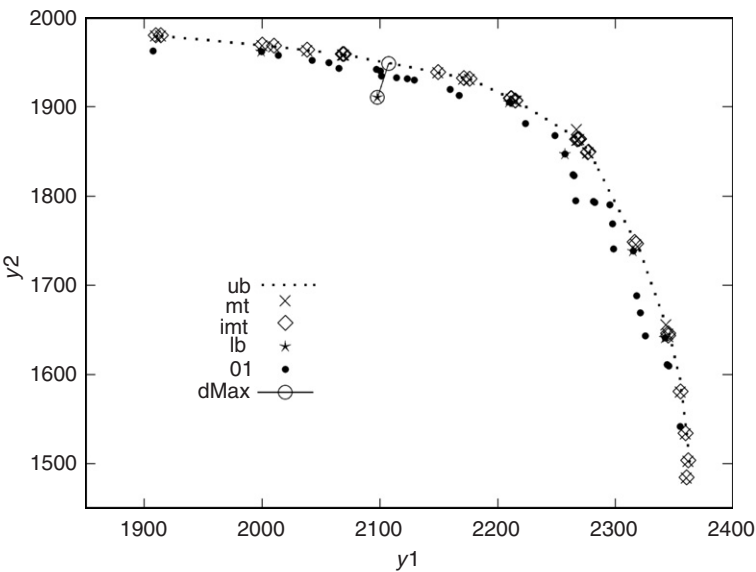


Fig. 4. Knapsack problem instance 50-1a.

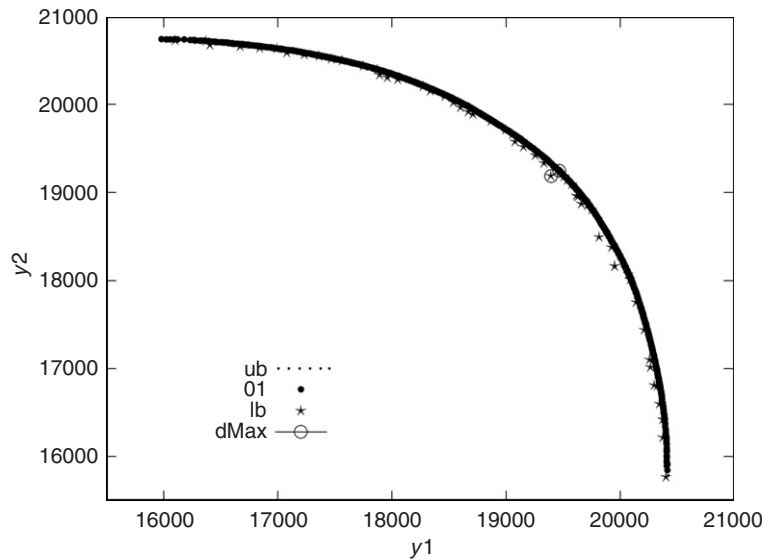


Fig. 5. Knapsack problem instance 500-1a.

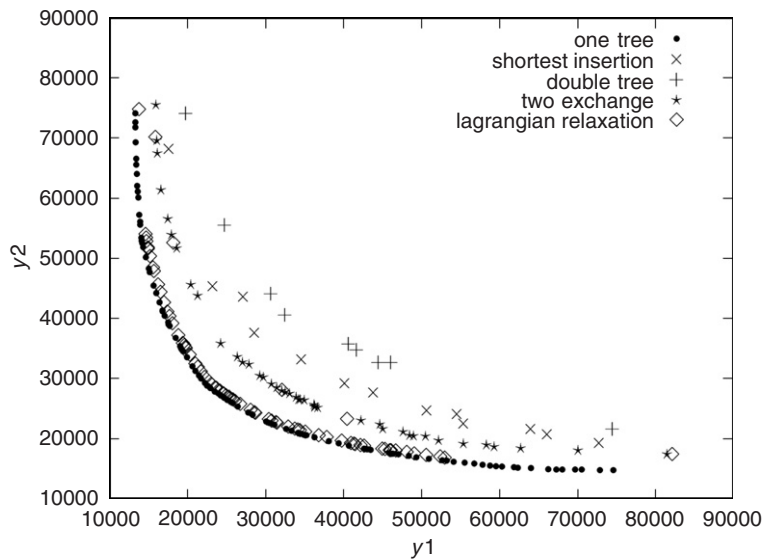


Fig. 6. All lower (one tree, Lagrangian relaxation) and upper (shortest insertion, double tree, two exchange) bound sets for TSP instance kroab50.

Note that, although the constraint matrix is totally unimodular nonsupported efficient solutions exist: total unimodularity only shows that feasible solutions correspond to extreme points of the polyhedron of the LP relaxation. It does not imply a similar statement for the convex hull of $Y = CX$. Because (P_λ) is polynomially solvable we used $L = (\text{conv } Y_{sN})_N$ and $U = Y_{sN}$ as described in Section 3.1.

Fig. 2 shows how L and U are constructed for a very small ($n = 5$) example, including the local nadir points for U . For these instances of the assignment problem the bound sets are very good, see Table 1 and Fig. 3 ($n = 50$) for a visual appreciation. The actual Y_N is shown for comparison.

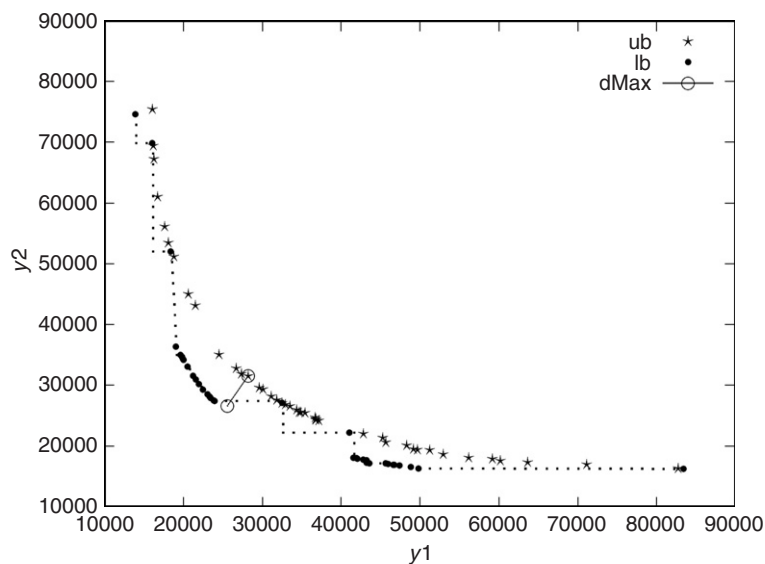


Fig. 7. Improved convex hull lower bound set (from Lagrangian relaxation) and combined upper bound set (from all upper bound sets in Fig. 6) for instance kroab50.

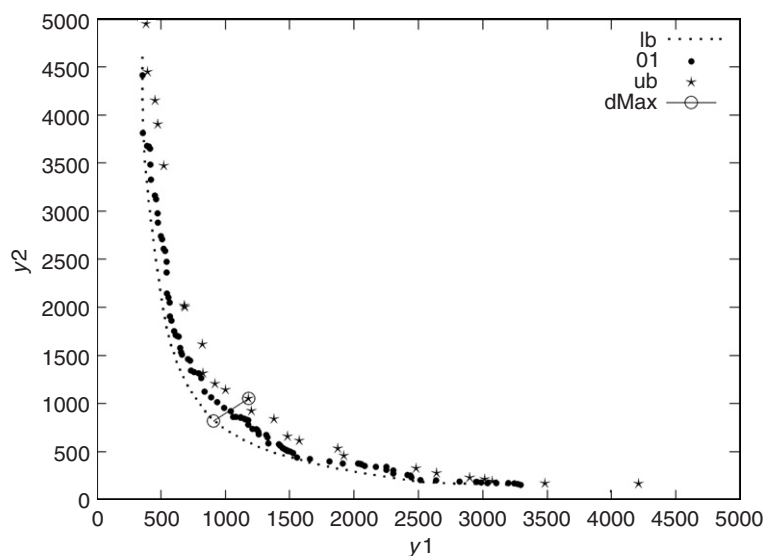


Fig. 8. Set covering instance 2scp62A.

4.2. The Knapsack problem

The bicriteria binary knapsack problem is the following integer programming problem.

$$\begin{aligned}
 \max \quad & \left(\sum_{i=1}^n c_i^1 x_i, \sum_{i=1}^n c_i^2 x_i \right) \\
 \text{s.t.} \quad & \sum w_i x_i \leq W, \\
 & x_i \in \{0, 1\}, \quad i = 1, \dots, n.
 \end{aligned}$$

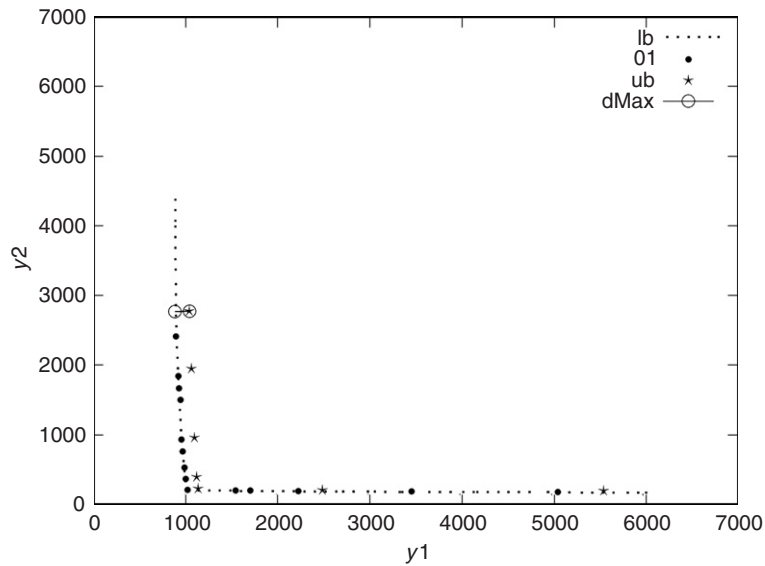


Fig. 9. Set covering instance 2scp81C.

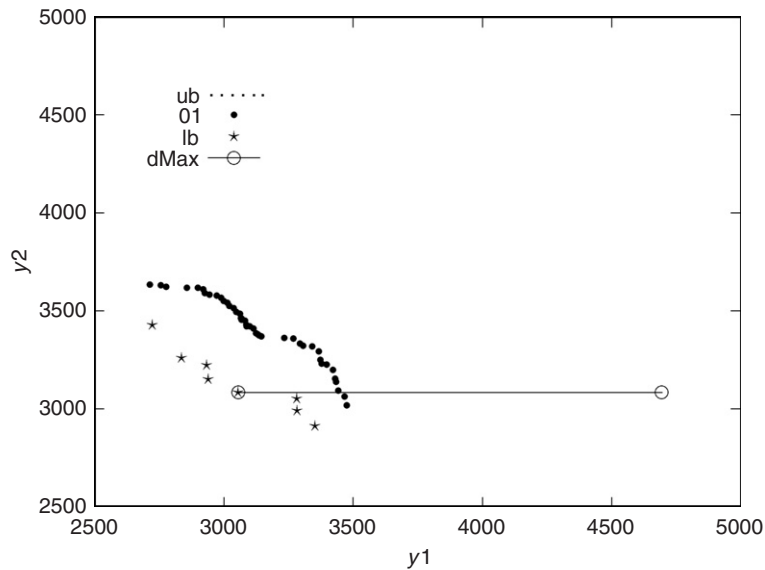


Fig. 10. Set packing: maximum independent set instance 2mis200_1000a.

We assume $0 < w_i < W$, $i = 1, \dots, n$; $\sum_{i=1}^n w_i > W$, and $c_i > 0$, $i = 1, \dots, n$. We will use the notation $c_i(\lambda) = \lambda c_i^1 + (1 - \lambda)c_i^2$, $0 \leq \lambda \leq 1$ and assume that

$$\frac{c_1(\lambda)}{w_1} \geq \frac{c_2(\lambda)}{w_2} \geq \dots \geq \frac{c_n(\lambda)}{w_n}.$$

Let $s = \min\{k : \sum_{i=1}^k w_i > W\}$ be the critical item. The three upper bounds for (P_λ) we shall use are the LP-bound, the Martello–Toth bound and the improved Martello–Toth bound, see [23]. The roles of lower bound sets and upper bound sets are of course interchanged here.

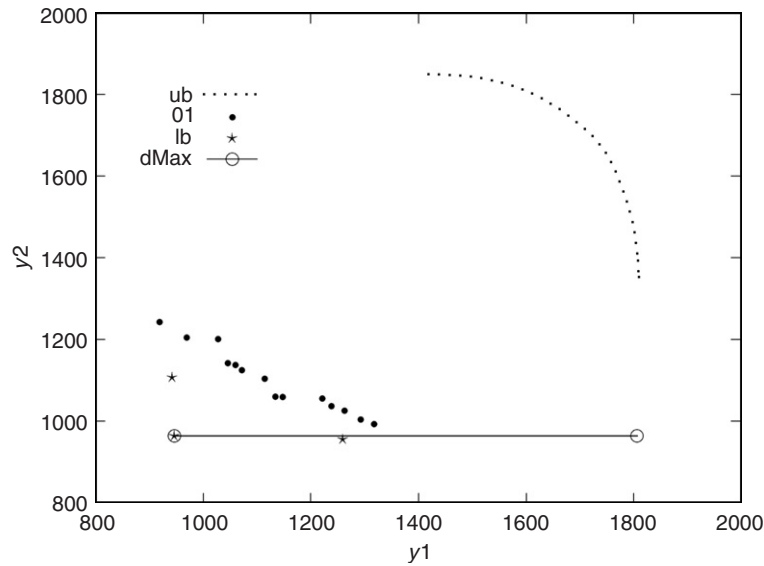


Fig. 11. Set packing instance 2spp203_600a.

The LP bound $U_{LP}(\lambda)$ is computed as the optimal solution of the LP relaxation of (P_λ) .

$$x_i^{LP} = \begin{cases} 1 & i = 1, \dots, s-1, \\ \frac{W - \sum_{i=1}^{s-1} w_i}{w_s} & i = s, \\ 0 & \text{else.} \end{cases} \quad (6)$$

Therefore, $U_{LP}(\lambda) = \sum_{i=1}^n c_i(\lambda)x_i^{LP}$. In fact, the bicriteria linear knapsack problem $\min\{Cx : w^T x \leq W; 0 \leq x_i \leq 1, i = 1, \dots, n\}$ is a particularly simple bicriteria LP, and from multicriteria linear programming, it is well known that all efficient solutions are obtained as solutions of weighted sum problems. Therefore, our bounding procedure computes the nondominated set of the linear relaxation of the bicriteria binary knapsack problem. Combining this with Proposition 1, we obtain Corollary 1.

Corollary 1. *The nondominated set of the bicriteria continuous knapsack problem is an upper bound set for the nondominated set of the bicriteria 0–1 knapsack problem.*

The Martello–Toth bound $U_{MT}(\lambda)$ [23, p. 20] improves upon the LP bound by discriminating the two mutually exclusive cases of item s being included in or excluded from the solution, thus using two solutions x^0 and x^1 .

$$x_i^0 = \begin{cases} 1 & i = 1, \dots, s-1, \\ \frac{W - \sum_{i=1}^{s-1} w_i}{w_{s+1}} & i = s+1, \\ 0 & \text{else,} \end{cases} \quad (7)$$

$$x_i^1 = \begin{cases} 1 & i = 1, \dots, s-2; \quad i = s, \\ \max \left\{ 0, 1 - \frac{w_s - (W - \sum_{i=1}^{s-1} w_i)}{w_{s-1}} \right\} & i = s-1, \\ 0 & \text{else.} \end{cases} \quad (8)$$

Hence, $U_{MT}(\lambda) = \max\{c(\lambda)x^0, c(\lambda)x^1\}$. Several researchers have further improved the bound by considering inclusion and exclusion of items before, respectively, after the critical item s , see [23, p. 22].

Let $\sigma^1(s) = \min\{k : \sum_{i=1}^k w_i > W - w_s\}$ and let $\sigma^0(s) = \min\{k : \sum_{i=1, i \neq s}^k w_i > W\}$. Define

$$x_i^0(s) = \begin{cases} 1 & i = 1, \dots, \sigma^0(s) - 1; \ i \neq s, \\ \frac{W - \sum_{i=1, i \neq s}^{\sigma^0(s)} w_i}{w_{\sigma^0(s)}} & i = \sigma^0(s), \\ 0 & \text{else,} \end{cases} \quad (9)$$

$$x_i^1(s) = \begin{cases} 1 & i = 1, \dots, \sigma^1(s) - 1; \ i = s, \\ \frac{W - w_s - \sum_{i=1}^{\sigma^1(s)-1} w_i}{w_{\sigma^1(s)}} & i = \sigma^1(s), \\ 0 & \text{else.} \end{cases} \quad (10)$$

and choose the x that gives the higher value of the objective, as before, to obtain $U_{\text{IMT}}(\lambda)$.

Some interesting observations can be made concerning the relationship of the various upper bounds. In the single objective case it is known that $U_{\text{IMT}}(\lambda) \leq U_{\text{MT}}(\lambda) \leq U_{\text{LP}}(\lambda)$ (see [23] for proofs). We shall show, by means of an example, that an analogous conclusion cannot be made in the biobjective case.

Example 1. Consider an instance of the bicriteria knapsack problem with $n = 8$, $W = 102$ and weights and values given as in the following table.

	1	2	3	4	5	6	7	8
c^1	15	100	90	60	40	15	10	1
c^2	15	100	90	x	40	15	10	1
w	2	20	20	30	40	30	60	10

Below we list the value to weight ratios for $\lambda = 1$, and the corresponding solutions x^{LP} , x^0 , and x^1 needed for the upper bounds.

$c(\lambda)/w$	1	2	3	4	5	6	7	8
	7.5	5	4.5	2	1	0.5	1/6	0.1
x^{LP}	1	1	1	1	0.75	0	0	0
x^0	1	1	1	1	0	1	0	0
x^1	1	1	1	2/3	1	0	0	0

Clearly, the $U_{\text{MT}}(\lambda)$ bound (obtained from x^1) is 285 and $U_{\text{LP}}(\lambda) = 295$. Computing objective values for these solutions we get $z(x^{\text{LP}}) = (295, 235 + x)$ and $z(x^1) = (285, 245 + (\frac{2}{3})x)$. Therefore, U_{MT} dominates U_{LP} if and only if $x \leq 30$.

We complement the presentation of upper bound sets for the knapsack problem with a simple lower bound set obtained by the greedy algorithm. The greedy lower bound $L_{\text{GR}}(\lambda)$ is given by either choosing all items before the critical one or the critical item alone, whichever is better.

$$x'_i = \begin{cases} 1 & i = 1, \dots, s-1, \\ 0 & \text{else,} \end{cases} \quad x''_i = \begin{cases} 1 & i = s, \\ 0 & \text{else.} \end{cases}$$

Then $L_{\text{GR}}(\lambda) = \max\{c(\lambda)x', c(\lambda)x''\}$ yields x_{GR} . In the single objective case, it is easily seen that the optimal solution is not greater than twice the greedy lower bound.

The quality of the bounds follows from their single objective performance ratios (2 in all cases) and Propositions 6 and 7.

In Fig. 4 ($n = 50$) we show the convex hull upper bound set obtained from LP relaxation, the points obtained using the Martello–Toth and improved Martello–Toth bounds for (P_λ) as well as lower bound points obtained from the greedy

heuristic. The larger instance ($n = 500$) shows the LP relaxation and greedy solutions. In general the results for the knapsack problem are the best we have obtained. Both figures show the true set of nondominated points for comparison.

4.3. The travelling salesperson problem

Our third example is the travelling salesperson problem. It is formulated as an integer programming problem here:

$$\begin{aligned} \min \quad & \left(\sum_{[i,j] \in E} c_{ij}^1 x_{ij}, \sum_{[i,j] \in E} c_{ij}^2 x_{ij} \right) \\ \text{s.t.} \quad & \sum_{j=1}^n x_{ij} = 2, \quad i = 1, \dots, n, \\ & \sum_{[i,j] \in E(C)} x_{ij} \leq |C| - 1, \quad C \subset E, \quad 3 \leq |C| \leq \left\lfloor \frac{n}{2} \right\rfloor, \end{aligned} \quad (11)$$

$$x_{ij} \in \{0, 1\}, \quad [i, j] \in E. \quad (12)$$

We assume that both distance matrices C^1 and C^2 are symmetric and satisfy the triangle inequality. We use Lagrangian relaxation of the degree constraints (11) for all nodes except 1 to obtain the following problem:

$$\begin{aligned} \min \quad & \sum_{[i,j] \in E} c_{ij}(\lambda) x_{ij} + \sum_{i=1}^n \pi_i \left(2 - \sum_{j=1}^n x_{ij} \right) \\ \text{s.t.} \quad & \sum_{j=1}^n x_{1j} = 2, \\ & \sum_{[i,j] \in E(C)} x_{ij} \leq |C| - 1, \quad C \subset E \setminus \{1\}, \\ & 3 \leq |C| \leq \left\lfloor \frac{n}{2} \right\rfloor \\ & x_{ij} \in \{0, 1\}, \quad [i, j] \in E. \end{aligned} \quad (13)$$

This problem is a minimal 1-tree problem with cost $c_{ij}(\lambda, \pi)$ depending on weight λ and multiplier π . For fixed λ and π values the optimal solution consists of a minimal spanning tree on nodes 2 to n plus the cheapest two edges incident on node 1 (thus a spanning tree plus one edge, a 1-tree). We solved (13) with $\pi = 0$ to get an initial bound and performed subgradient optimization as described in [24] to improve the original 1-tree bound. Let the bound obtained with $\pi = 0$ be $z_{LB}(\lambda)$. Let $z_{UB}(\lambda)$ be the value of an upper bound, i.e. a feasible tour, which we computed using a 2-opt heuristic with 30 random restarts. Let $G_i = 2 - \sum_{j=1}^n x_{ij}$, let

$$T = \frac{\delta(z_{UB} - z_{LB})}{\sum_{i=1}^n G_i^2}$$

be a step length, and define new π values as $\pi_i := \pi_i + T G_i$. We set $\delta = 2$ initially and allowed 10 iterations without improvement in the lower bound value before halving δ . The procedure stops if either a tour is obtained or δ falls below 0.01.

If the procedure stops with a tour, we have identified a supported efficient solution of TSP. If, however, the procedure stops with a 1-tree that is not a tour, the lower bound value (the objective value of the 1-tree problem (13)) contains the nonzero component $2 \sum_{i=1}^n \pi_i - \sum_{i=1}^n \sum_{j=1}^n (\pi_i + \pi_j) x_{ij}$. Since this cannot be decomposed into the two objectives, we cannot use the corresponding 1-tree to compute a point in the lower bound set for the biobjective problem. In this case we revert to the original 1-tree obtained with $\pi = 0$. Therefore, we will in general end up with nonempty S_2 in our bounding procedure and therefore use the improved convex hull lower bound set L' .

For the upper bounding procedure (see Section 3.2) we used the tree heuristic, cheapest insertion, and 2-opt with random restart (30 restarts). See e.g. [25] for a more detailed description of these heuristics.

In Fig. 6 we present all of the upper bound sets obtained by the different heuristics for an instance with 50 cities. Clearly, as in the single objective case, two-exchange dominates shortest insertion, which in turn dominates the tree heuristic. The figure also shows that Lagrangian relaxation improves the initial 1-tree very often, occasionally finding a supported feasible solution. Fig. 7 illustrates how the improved convex hull bound set is constructed, and how several upper bound sets are compounded (same instance). In all TSP instances there is a clear gap between L and U . The true set Y_N is unknown for the TSP instances.

4.4. The set covering and set packing problems

Our final examples are the set covering and set packing problems. They are known to be not only theoretically but also computationally very difficult. Our results provide another confirmation of this observation. First, we state the binary programming formulation.

$$\begin{aligned} \min(\max) \quad & \left(\sum_{i=1}^n c_i^1 x_i, \sum_{i=1}^n c_i^2 x_i \right) \\ \text{s.t.} \quad & \sum_{i=1}^n a_{ji} x_i \geq (\leq) 1 \quad j = 1, \dots, m, \\ & x_i \in \{0, 1\} \quad i = 1, \dots, n. \end{aligned}$$

The coefficients are $a_{ji} \in \{0, 1\}$ and $a_{ji} = 1$ means variable x_i “covers” constraint j . For set covering we compute lower bounds for the weighted sum problems through the continuous relaxation. For upper bounds we chose a greedy strategy, where the next variable to be set to 1 is given by the smallest $c_i(\lambda)$ such that x_i covers an additional constraint. The procedure stops when all constraints are satisfied.

For set packing the upper bound set is computed using the LP relaxation, the lower bound set using a greedy heuristic, where the variable with lowest $c_i(\lambda)$ is added, as long as it does not violate any constraint. In addition a local search exchange heuristic is performed on the result of the greedy heuristic. Figs. 8 and 9 show that the bound sets give a good idea where the nondominated points are located. But this problem, as well may have very unusual “shapes” of the nondominated set.

The set packing problem serves as example that simple heuristics and relaxations may yield bound sets that are very far away from the true nondominated set (yet still better than the ideal point, but sometimes even much worse than the nadir point). For both problems the true set Y_N is shown.

5. Conclusions

We have introduced the concept of bound sets for biobjective combinatorial optimization problems. Motivated by the fact that (MOCO) problems are very difficult in general, yet appear often in practical applications, we duplicated the transition from one optimal solution in single objective optimization to a set of efficient solutions in the multicriteria case. We gave a general scheme for computing a lower bound set and proved some general results. With numerical tests we illustrated that good bounds can be calculated for some problems (AP, KP, TSP) whereas simple procedures such as LP relaxation and greedy heuristics yield poor results on other problems (SPP, SCP). We observe that the quality of bound sets is often dependent on characteristics of the two objective functions (see SPP). Note, however, that these results are not intended as an extensive numerical study of the quality of relaxation and heuristic methods but rather as qualitative evidence that the concept of bound sets is meaningful and that its use can have advantages over the use of ideal and nadir points of some sort (true, local, of a relaxation) e.g. in multiobjective branch-and-bound algorithms.

Further research in this area can be very fruitful for the development of algorithms that allow a more efficient solution of MOCO problems than possible today. Our investigation has been restricted to bound sets based on weighted sum scalarization. Theoretical results and empirical studies show that this approach has limits. It is therefore worthwhile studying bounds based on other scalarizations, in particular the ε -constraint method [26], which can be used to find nonsupported efficient solutions.

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